

Dina Abdel Rahman

Lead Data, AI & Cloud Architect

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Professional Summary

Experienced Data, AI and Cloud Architect with 19+ years of experience delivering complex data, analytics, AI and cloud solutions across public sector, energy, utilities, telecom and banking. Strong background in enterprise architecture, applied AI solution design, GenAI-enabled platforms, Claude-based solutions, cloud-native systems, data governance and large-scale transformation. Experienced in leading customer-facing solution design from discovery and assessment through architecture, MVP/POC delivery, stakeholder alignment and production readiness. Strong ability to translate complex AI and data capabilities into practical, secure and scalable business solutions across regulated and enterprise environments.

Skills

Applied AI, GenAI & LLM Architecture:

- Designed enterprise GenAI platforms using Claude and OpenAI, covering AI agents, orchestration, knowledge ingestion, action execution and multi-agent coordination.
- Developed Claude and OpenAI-based MVPs for banking and energy sector use cases, supporting automation, knowledge discovery, data analysis and decision support.
- Designed AI agent solutions using Claude, OpenAI, Microsoft Agent Framework, LangChain and Azure AI services.
- Architected secure GenAI solutions covering model access, prompt workflows, enterprise data integration, output validation, monitoring and governance controls.
- Supported AI MVP and POC delivery by translating business problems into practical use cases, solution patterns and implementation plans.
- Applied LLM evaluation, prompt design, retrieval patterns and controlled access to improve reliability, safety and business usability of AI solutions.

Data Engineering, Modelling & Integration:

- Designed enterprise data models, warehouse structures and analytical data layers to support reporting, analytics and operational intelligence.
- Designed big data and distributed data platform architectures using batch, streaming and historical datasets.
- Architected data ingestion and integration patterns across enterprise systems, APIs, files, logs, databases and event streams.
- Designed data streaming architectures using Kafka and event-driven patterns to support near-real-time data processing.
- Applied data security, access control, governance and quality controls to support trusted, auditable and compliant data use.
- Supported data platform modernisation covering data modelling, normalisation, schema design, storage selection and integration architecture.

Architecture, Governance & Leadership:

- Acted as architecture lead and design authority on complex programmes, facilitating architecture reviews, option assessments and decision forums.
- Led architecture assessment and assurance activities, validating solution designs against enterprise principles, security standards and operational requirements.
- Advised senior stakeholders on technology evaluation and buy-vs-build decisions, balancing innovation with risk, cost and regulatory constraints.
- Supported Agile delivery, collaborating with development teams on backlog refinement, sprint planning, solution validation and release coordination.
- Produced clear, structured architecture documentation and executive-level reporting, enabling effective governance, knowledge transfer and long-term platform evolution.

Migration & Large-Scale Transformation

- Led and supported end-to-end data and platform transformation programmes, modernising legacy and fragmented data landscapes into scalable, cloud-native and hybrid architectures.
- Designed enterprise ingestion and integration architectures onboarding large numbers of heterogeneous data sources, spanning relational databases, NoSQL stores, files, APIs, logs and event streams.
- Architected and governed ETL/ELT pipelines and ingestion frameworks enabling reliable batch and near-real-time data processing at scale.
- Delivered migration-critical design activities including schema rationalisation, data normalisation, dependency analysis, and data quality enforcement, ensuring continuity of service and regulatory compliance.
- Defined backup, disaster-recovery and high-availability strategies aligned to SLA and operational resilience requirements for mission-critical data platforms.
- Produced target-state architectures and phased delivery roadmaps, balancing technical debt reduction, business value and delivery risk in complex enterprise environments.

Tools and Platforms:

- **AI, GenAI & Agentic Platforms:** Claude, OpenAI, Azure AI Foundry, Azure ML Studio, Microsoft Agent Framework, LangChain, LangGraph, CrewAI, SageMaker.
- **Programming & API Development:** Python, FastAPI, Flask, scikit-learn, PyTorch.
- **Data Platforms:** Oracle, MySQL, PostgreSQL, Cassandra, Neo4j, Victoria Metrics, Redis, Elasticsearch, Teradata, Cloudera, Apache Spark, Apache Kafka, Apache Airflow, Microsoft Fabric, AWS Glue, AWS Athena, Alteryx.
- **Analytics & Visualisation:** Power BI, MicroStrategy, Tableau.
- **DevOps & Cloud Native:** Git, Jenkins, GitLab, Azure DevOps, Terraform, Docker, Kubernetes, OpenShift.

Experience:

Data & AI Solution Architect – Associate Director | KPMG | Ireland (May 2025 – Present):

- Deliver Data, AI and cloud solution architecture services across public sector, utilities, energy and banking engagements.
- Lead solution design across data architecture, data management, cloud modernisation, governance, security and AI-enabled platforms.
- Led customer-facing AI workshops and technical discussions to shape use cases, validate solution options and support MVP/POC delivery.
- Support clients across assessment, target-state architecture, solution options, delivery planning and architecture assurance.
- Work with business, data, IT, security and delivery teams to translate requirements into practical and scalable architecture designs.
- Contribute to tenders and business development activities, including solution design, delivery planning and resource estimation.

Main Projects

Claude-Based Enterprise AI Platform | Banking Client

- Delivered architecture design for a Claude-based enterprise AI platform deployed on Azure for a regulated banking client.
- Defined the end-to-end platform architecture covering Claude integration, AI agents, orchestration, knowledge ingestion, data integration, governance and operational tooling.
- Designed core platform modules, including agent runtime, action execution, multi-agent coordination and secure access to enterprise knowledge sources.
- Designed secure cloud infrastructure covering networking, data storage, model access, monitoring, observability and environment separation.
- Embedded security, cost management, auditability and operational controls to support safe, scalable and governed AI adoption.

Sustainability Data & Technology Assessment | Energy Client

- Led a sustainability data and technology assessment for a large, regulated energy organisation.
- Reviewed the end-to-end data, technology and governance landscape supporting ESG and CSRD reporting.
- Assessed the existing sustainability reporting solution across performance, data quality, security, auditability and integration.

- Analysed ESG data flows across enterprise systems, analytics platforms, geospatial tools and spreadsheet-based processes.
- Led a buy-versus-build assessment and defined future-state architecture principles, data controls and ESG lifecycle requirements.

Automated Invoice Processing Architecture | Public Sector Client

- Designed Azure-based cloud architecture for automated invoice processing.
- Designed the end-to-end invoice intake and validation process using scanned invoice images.
- Used Azure Document Intelligence to support invoice classification and OCR-based data extraction.
- Defined the architecture for secure storage of extracted invoice data, processing artifacts and validation results.
- Designed security controls to protect invoice data, scanned documents and system-generated artifacts.

Core Data Platform Assessment | Public Sector Client

- Performed an end-to-end assessment of the client's core data system and supporting infrastructure.
- Reviewed database models, table structures, constraints, storage design, internal procedures and bulk load processing.
- Identified archive requirements and helped define the archive process and target archive solution.
- Assessed data quality issues across critical data elements and established data quality contract guidelines.
- Reviewed authentication, authorisation, security logging, monitoring, backup, high availability and disaster recovery arrangements.
- Recommended improvements to support service continuity, restoration, resilience and agreed SLA requirements.

Claude MVP Development | Energy Sector

- Developed five Claude-based MVPs for energy sector use cases, supporting applied AI adoption across business and operational processes.
- Designed MVP architecture covering prompt workflows, enterprise data access, user interaction flows, security controls and output validation.
- Translated business problems into practical GenAI use cases, solution patterns and reusable MVP components.
- Worked with stakeholders to validate use case value, technical feasibility and potential delivery approach.
- Supported safe and governed AI adoption by considering data access, privacy, reliability, auditability and operational controls.

Senior Solution Architect | Ericsson | Ireland (Nov 2022 – May 2025):

- Led the architecture design for the ESOA assurance sub-system, a next generation 5G network automation platform within the BSS landscape.
- Drove technical discussions with product, engineering and stakeholder teams to analyse requirements, risks, dependencies and solution options.
- Designed scalable cloud-native components, modular services and data flow diagrams for a distributed microservices-based environment.
- Assessed solution designs covering microservices, data stores, message bus integration, API specifications and platform dependencies.
- Designed and assessed data transformation flows, ETL logic, topic message schemas and API payload structures.
- Contributed to ML and Generative AI use cases, supporting solution design and implementation guidance for product efficiency improvements.
- Reviewed user management, monitoring, maintenance and operational requirements to support platform reliability and service assurance.
- Analysed integration points and aligned solution dependencies with internal teams and external counterparts.
- Produced technical solution documentation and reviewed designs with Product Managers, Product Owners and engineering teams.
- Supported development and testing teams through weekly planning, solution clarification and delivery alignment.

AI, Data Solution Architect | DU | UAE (May 2019 – Nov 2022)

- Led the architecture and implementation of DU's enterprise data lake, supporting large-scale ingestion, governance, analytics and AI use cases.
- Assessed and integrated 60+ data sources across RAN, core, transmission, IP, applications, probes, infrastructure and operational systems.

- Designed the ingestion architecture covering source connectivity, data structures, data frequency, normalisation, field extraction and data quality controls.
- Designed Kafka platform components, including topic structure, data schema, retention policy, compression and security controls.
- Designed Cloudera platform components, including HDFS structure, data formats, Spark, Hive, Impala jobs and Ranger security policies.
- Led AI and data project assessments covering use case identification, complexity analysis, data availability, ROI and delivery planning.
- Evaluated platforms for model training, inference and lifecycle management, supporting AI solution selection and implementation planning.
- Prepared POCs for AI, analytics and automation use cases, using stakeholder feedback to move concepts toward functional delivery.
- Managed engagement with business, technology, security, infrastructure, vendors, government entities and external partners.
- Led RFI/RFP activities for network assurance and automation platforms, including use case definition, vendor evaluation and scoring.
- Designed IoT and analytics solution proposals for telecom field operations, fleet management, industrial IoT and patrol management use cases.

Data Analytics and Engineering Lead | DU | UAE (August 2014 – May 2019)

- Led data analytics and engineering activities across network operations, performance management and operational reporting.
- Extracted, integrated and analysed high-volume data from probing systems, network platforms and operational systems.
- Developed SQL scripts, ETL logic and reporting datasets to support network investment, rollout planning and feature deployment decisions.
- Built real-time performance dashboards with low-latency updates to support operational monitoring and service assurance.
- Prepared network traffic forecasting and capacity planning analysis using device, coverage, technology and usage data.
- Analysed M2M and enterprise device performance to identify service trends, issues and optimisation opportunities.
- Represented network operations in discussions with business, marketing and senior stakeholders.
- Led a multi-year network upgrade programme across operations, coordinating 15+ stakeholders and 40+ team members across RAN, core, transport and infrastructure teams.

Migration Consultant | Huawei | Various Locations (November 2007 – July 2014)

- Delivered reporting and performance monitoring solutions across telecom network transformation programmes.
- Designed real-time dashboards and KPI reporting solutions to improve visibility of network performance and operational trends.
- Extracted, integrated and analysed high-volume network and operational data from multiple source systems.
- Developed structured reporting workflows to provide accurate performance metrics for operations teams and leadership.
- Analysed network performance data to identify trends, anomalies and optimisation opportunities.
- Supported cross-functional stakeholder coordination, ensuring consistent delivery of dashboards, reports and data insights.

Professional Training

- Kubernetes and Helm Administration
- Developing Cloud Native Applications
- Microsoft Azure DevOps Engineer
- Microsoft Azure AI Engineer
- Microsoft Azure Machine Learning Engineer
- Microsoft Azure Solution Architect Professional
- Microsoft Azure Data Engineering
- Developing on AWS Professional Training

- DevOps Engineering on AWS Professional Training
- AWS Machine Learning Engineer Professional Training
- Intel® Edge AI for IoT Developers

Education

BSc Electronics and Communication Engineering | Cairo University (2002 – 2007) General grade very good

Certifications

Microsoft Certified: Azure AI Engineer Associate | Microsoft (May 2026)

Microsoft Certified: Azure Data Fundamentals | Microsoft (March 2022)

Microsoft Certified: Azure AI Fundamentals | Microsoft (March 2022)

Microsoft Certified: Azure Fundamentals | Microsoft (Feb 2022)

AWS Certified Machine Learning – Specialty | Amazon Web Services (Dec 2021)

AWS Certified Solution Architect Associate | Amazon Web Services (Oct 2021)

AWS Partner: Accreditation (Business) | Amazon Web Services (Dec 2021)

AWS Partner: Accreditation (Technical) | Amazon Web Services (Dec 2021)

AWS Partner: Cloud Economics Accreditation | Amazon Web Services (Dec 2021)